Addressing Mechanisms and Heterogeneity in Change with Usable Tools

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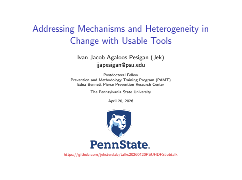
https://github.com/jeksterslab/talks20260420PSUHDFSJobtalk



- Good morning, and thank you for the opportunity to be here.
- I'm Jek, and my research program is about dynamic processes in development, family life, and health:
  - how change unfolds over time,
  - how people differ in those processes, and
  - how we can build tools that make those questions easier to study well.
- What draws me to HDFS is that these questions are inherently multilevel and contextual. They show up in people, dyads, families, and daily-life settings, and they increasingly require methods that can handle heterogeneity without losing substantive meaning.
- In this talk, I'll walk through three studies that reflect that agenda.

Training and mentorship

The Ateneo de Manila University | University of Macau | The Pennsylvania State University

Liane Peña-Alampay

Shu Fai Cheung

Sy-Miin Chow

Michael A. Russell

My work brings together **developmental science**, **quantitative methods**, and **dynamic modeling**, with an emphasis on usable methods.

- Before I get into the studies, let me briefly situate the kind of work I do. My training moved across developmental psychology, quantitative methods, and dynamic modeling.
  - At Ateneo, I built a strong developmental foundation.
    - I also taught undergraduate and graduate courses in statistics, psychometrics (classical test-theory and IRT), and General Psychology.
  - In Macau, I deepened my work in SEM, meta-analysis, and computation.
  - And here at Penn State, through PAMT, I extended that work into dynamic models and intensive longitudinal data.
- Together, those experiences shaped the research program I want to build: methods for dynamic psychological processes that are also efficient, reproducible, and genuinely usable for substantive researchers.

An intuition for inertia

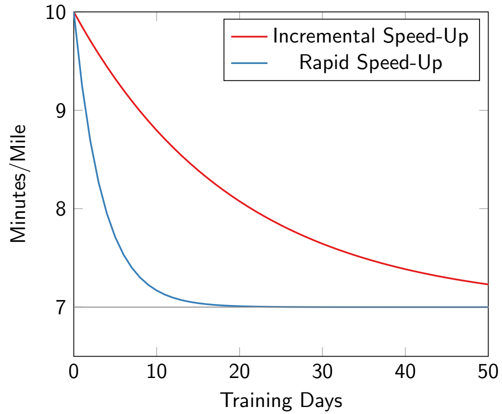
Inertia: how much yesterday still shapes today.

An intuition for inertia

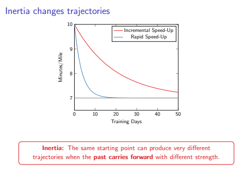
Inertia: how much yesterday still shapes today.

- So before getting into the three studies, I want to start with one simple intuition that connects all of them.

Inertia changes trajectories

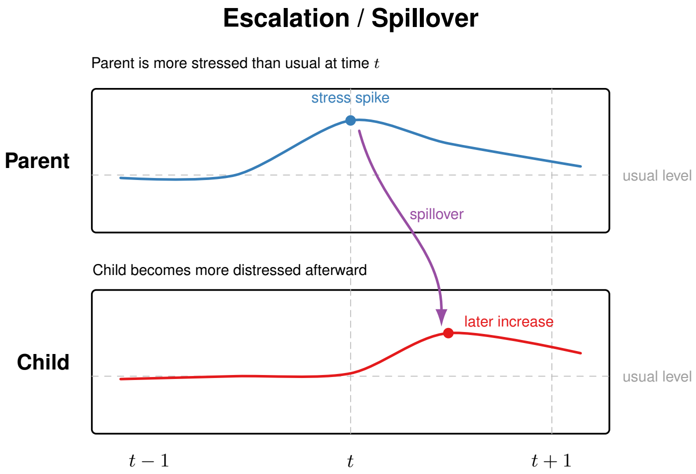


Inertia: The same starting point can produce very different trajectories when the **past carries forward** with different strength.

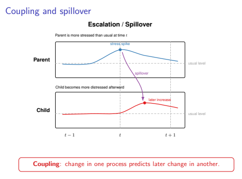


- Here both runners start at the same place, but they diverge over time. The difference is not where they began. It is how strongly earlier states carry forward.
- That is the key idea I want to carry into the talk: dynamic processes are shaped not only by level, but by persistence.
  - The stronger the inertia, the slower the process reaches the long term equilibrium.
  - The weaker the inertia, the faster the process reaches the long term equilibrium.
- So when I talk later about mediation, alcohol intoxication, or affect dynamics, this is the intuition underneath all of it.
- And in HDFS, that kind of persistence shows up in things like affective recovery, stress carryover, sleep, health behavior, and other daily-life processes.
- But dynamic processes are not only self-persistent. They can also shape one another.

Coupling and spillover



Coupling: change in one process predicts later change in another.



- That brings us to coupling. Here, a parent's stress spike is followed by greater child distress. So now we move from persistence within one process to spillover across connected processes.
- This is another core idea for the talk: dynamic systems are not just about how a variable carries itself forward, but also how change in one process predicts later change in another.
- And that sets up the three studies. Study 1 asks how dynamic effects unfold across time. Study 2 asks how different dynamic patterns emerge. And Study 3 asks how person-specific dynamics can be synthesized into population-level inference.
- That is exactly the kind of question HDFS asks all the time—not just what persists within a person, but how stress, affect, and regulation move across people and relationships.

Study 1

Psychological Methods

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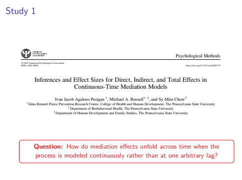
Inferences and Effect Sizes for Direct, Indirect, and Total Effects in Continuous-Time Mediation Models

Ivan Jacob Agaloos Pesigan [1], Michael A. Russell[1, 2], and Sy-Miin Chow[3]
[1] Edna Bennett Pierce Prevention Research Center, College of Health and Human Development, The Pennsylvania State University
[2] Department of Biobehavioral Health, The Pennsylvania State University
[3] Department of Human Development and Family Studies, The Pennsylvania State University

Question: How do mediation effects unfold across time when the process is modeled continuously rather than at one arbitrary lag?



- I'll start with Study 1, where the core question is how mediation effects unfold when we treat the process as continuous rather than tying it to one arbitrary lag.

Mediation is everywhere in HDFS and related fields.

- **Substantive** (e.g.; Bernardo et al., 2017; Pesigan et al., 2014; Zhang et al., 2020)
- **Methodological** (e.g.; Cheung & Pesigan, 2023b; Cheung et al., 2023; Pesigan & Cheung, 2020, 2024; Pesigan et al., 2023).
  - https://CRAN.R-project.org/package=semmcci
  - https://CRAN.R-project.org/package=semlbci
  - https://CRAN.R-project.org/package=betaDelta
  - https://CRAN.R-project.org/package=betaDSandwich

Study 1 extends that line of work into continuous time by adding **standardized effect sizes**, principled **uncertainty quantification**, and usable tools in `cTMed`.

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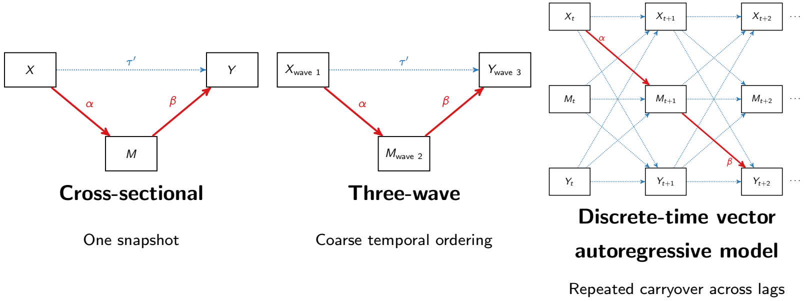
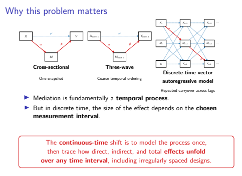
- I have worked on mediation from both the substantive and methods sides:
  - On the substantive side, my work has examined mechanisms such as the mediating role of drinking motivations in hazardous drinking.
  - On the methods side, I have worked on inference for indirect effects and related quantities through SEM-based confidence intervals, Monte Carlo confidence intervals with missing data, likelihood-based intervals in semlbci, bootstrap tools, delta method, and standardized effect-size inference.
- Study 1 extends that program into continuous time. Prior continuous-time mediation work established how direct, indirect, and total effects unfold across time (Deboeck & Preacher, 2015; Ryan & Hamaker, 2021).

Why this problem matters



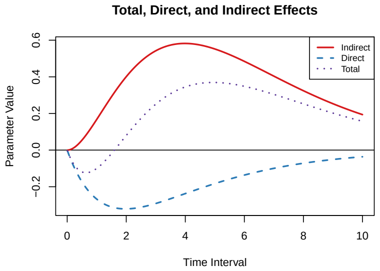
- Mediation is fundamentally a **temporal process**.
- But in discrete time, the size of the effect depends on the **chosen measurement interval**.

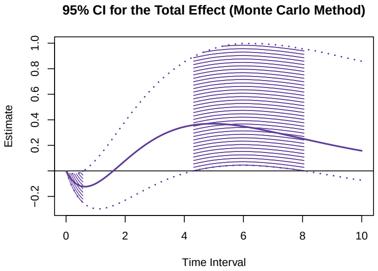
The **continuous-time** shift is to model the process once, then trace how direct, indirect, and total **effects unfold over any time interval**, including irregularly spaced designs.



- This slide gets at why the problem matters. Mediation is fundamentally a temporal process, but a lot of our traditional models only give us a few snapshots of that process. Cross-sectional models give us one snapshot, three-wave models give us coarse temporal ordering, and even discrete-time autoregressive models still depend on the particular measurement interval we happened to choose.
- The key shift in continuous time is that we model the underlying process once, and then trace how direct, indirect, and total effects unfold over any time interval. That gives us a way to study mediation as a genuinely dynamic process, including in designs with irregular spacing.
- For HDFS researchers, the payoff is that mechanisms in development, family process, and health behavior rarely operate at one arbitrary lag, and this framework lets us study

Novel contributions





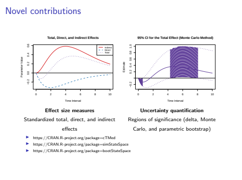
Effect size measures

Standardized total, direct, and indirect effects

Uncertainty quantification

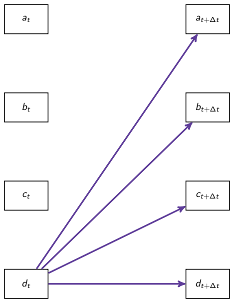
Regions of significance (delta, Monte Carlo, and parametric bootstrap)

- https://CRAN.R-project.org/package=cTMed
- https://CRAN.R-project.org/package=simStateSpace
- https://CRAN.R-project.org/package=bootStateSpace



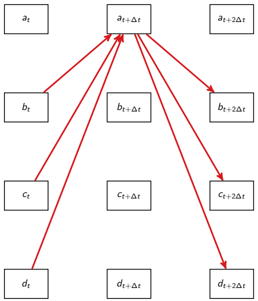
- This slide highlights the main contributions of Study 1.
  - First, I introduce standardized total, direct, and indirect effects for continuous-time mediation, so the results are scale-free and easier to interpret across studies.
  - Second, I develop a principled inference framework for uncertainty quantification, using delta, Monte Carlo, and parametric bootstrap methods to obtain standard errors, confidence intervals, and regions of significance across time.
  - Third, I translate those ideas into computational and visualization tools in the open-source cTMed package, so the methods are not just theoretically available, but practically usable.
- Talk briefly about the delta, Monte Carlo, and parametric bootstrap methods.

From mediation effects to system-level network summaries



Total effect centrality

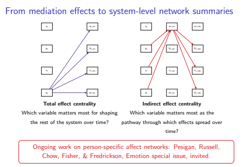
Which variable matters most for shaping the rest of the system over time?



Indirect effect centrality

Which variable matters most as the pathway through which effects spread over time?

Ongoing work on person-specific affect networks: Pesigan, Russell, Chow, Fisher, & Fredrickson, Emotion special issue, invited.



- Up to this point, I have framed mediation one pathway at a time. But many HDFS questions are really about systems, not isolated paths. Once we can estimate total, direct, and indirect effects across time intervals, we can ask a new question: which variables matter most in the whole system?
- Total effect centrality asks which variable has the **broadest downstream influence**, and indirect effect centrality asks which variable acts most as a **conduit through which effects spread**. So this moves the framework from single mechanisms to leverage points in a dynamic system.
- This is one direction I am pursuing now in ongoing work on affect networks during meditation training. The idea is to estimate person-specific affect dynamics across baseline, intervention, and follow-up, then summarize those systems with continuous-time total-effect and indirect-effect centrality.
- That lets us ask which emotions become leverage points in a person's system, whether those leverage points shift as training unfolds, and whether that reorganization relates to practice dose and later well-being.

Study 2

AMERICAN PSYCHOLOGICAL ASSOCIATION

Psychology of Addictive Behaviors

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Common and Unique Latent Transition Analysis (CULTA) as a Way to Examine the Trait–State Dynamics of Alcohol Intoxication

Ivan Jacob Agaloos Pesigan[1], Michael A. Russell[1, 2], and Sy-Miin Chow[3]
[1] Edna Bennett Pierce Prevention Research Center, College of Health and Human Development, The Pennsylvania State University
[2] Department of Biobehavioral Health, The Pennsylvania State University
[3] Department of Human Development and Family Studies, The Pennsylvania State University

Question: Can inertia help distinguish qualitatively different drinking patterns, beyond differences in average intoxication level?

Study 1 focused on how dynamic effects unfold across time. Study 2 asks a different but related question: can inertia itself help define qualitatively different risk patterns? This is where I move from quantifying dynamic effects to building a new latent-variable framework for decomposing heterogeneity in intensive longitudinal alcohol data.

From addiction research to dynamic modeling

Why Study 2 is exciting

- Addiction has been a consistent substantive thread in my work (Herring et al., 2016; Russell et al., 2025; Yu et al., 2015, 2019; Zamboanga, Iwamoto, et al., 2015; Zamboanga, Pesigan, et al., 2015; Zhang et al., 2019, 2020).
- **Substantive question:** How do risky drinking patterns persist or change across days?
- **Methodological challenge:** Transdermal alcohol concentration (TAC) features are multidimensional, partly shared, partly feature-specific, and dynamic over time.

Substantive grounding in addiction research, together with the quantitative expertise to build **new models** when the science demands them.

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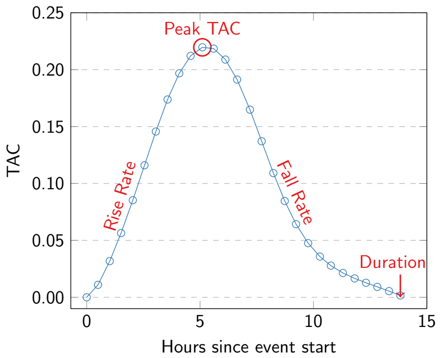
Methodological challenge: Transdermal alcohol concentration (TAC) features are multidimensional, partly shared, partly feature-specific, and dynamic over time.
Substantive grounding in addiction research, together with the quantitative expertise to build new models when the science demands them.

- This study is a natural next step for me. Addiction and alcohol have been a consistent part of my substantive work, but Study 2 is where that background meets the kind of quantitative contribution I want to make in HDFS. TAC sensors give us richer, more objective data—but they also create a harder modeling problem, because these features are partly shared, partly feature-specific, and changing across days. That is exactly the setting that motivated CULTA.
- Richer alcohol data created a harder modeling problem, and that is what motivated CULTA.

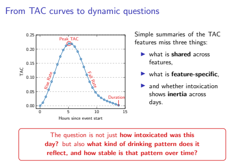
From TAC curves to dynamic questions



Simple summaries of the TAC features miss three things:

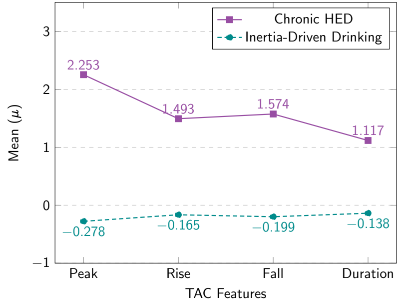
- what is **shared** across features,
- what is **feature-specific**,
- and whether intoxication shows **inertia** across days.

The question is not just **how intoxicated was this day?** but also **what kind of drinking pattern does it reflect, and how stable is that pattern over time?**

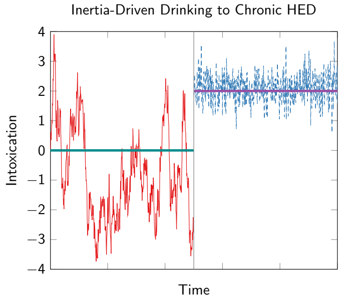


- A TAC curve can be summarized by peak, rise rate, fall rate, and duration. But once we do that, a new question appears: are we looking at one common intoxication process, feature-specific expressions of intoxication, or both? And beyond that, does intoxication carry over across days? So the substantive question becomes not just how intoxicated this day was, but what kind of drinking pattern it reflects and how stable that pattern is over time.
- That is exactly the problem CULTA was designed to solve.

Profiles differed in mean intoxication and inertia



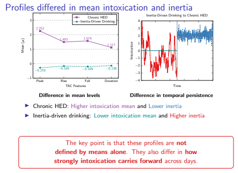
Difference in mean levels



Difference in temporal persistence

- Chronic HED: Higher intoxication mean and Lower inertia
- Inertia-driven drinking: Lower intoxication mean and Higher inertia

The key point is that these profiles are **not defined by means alone**. They also differ in **how strongly intoxication carries forward** across days.



- CULTA answers that question by separating overall intoxication level from temporal persistence.
- The main empirical result is that two profiles emerged, but what matters is how they differ. Chronic HED is characterized by higher mean intoxication and lower inertia, while inertia-driven drinking shows lower mean intoxication but stronger carryover across days.
- So this is not just a classification result. It is a decomposition of heterogeneity showing that risky drinking patterns differ not only in level, but in temporal persistence.

AUD risk predicts getting stuck in chronic HED

Start in chronic HED	**Stay in chronic HED**	**Shift into chronic HED**

CULTA does not just classify drinking days; it helps identify **who is likely to get stuck in riskier dynamic patterns**.

AUD risk predicts getting stuck in chronic HED
Start in chronic HED
Stay in chronic HED
Shift into chronic HED
CULTA does not just classify drinking days; it helps identify who is likely to get stuck in riskier dynamic patterns.

- AUDIT helps identify who is more likely to start in, remain in, or shift into the chronic HED profile. So the model is not only descriptive—it has screening value. And beyond classification, one of the paper's practical implications is that duration and fall rate appear to be more promising intervention targets than peak or rise rate, because they showed more substantial person-specific variability.
- So Study 2 showed that heterogeneity in dynamics is not just about mean level. It is also about temporal persistence, profile membership, and who gets stuck in riskier patterns. Once we start treating those person-specific dynamic signatures as substantive targets, the next question becomes: how do we synthesize them at the population level without averaging them away?
- More broadly, this is an HDES problem because many risk

Study 3

Under review at *Multivariate Behavioral Research*

From Person-Specific Dynamics to Population Inference: A Computationally Efficient Meta-Analytic Structural Equation Modeling Framework for Integrating Complex Multilevel Dynamic Models

Ivan Jacob Agaloos Pesigan[1], Michael A. Russell[1, 2], and Sy-Miin Chow[3]
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Question: How can we draw **population-level** conclusions from **person-specific dynamics** without averaging away the very individual differences we care about?

Study 3
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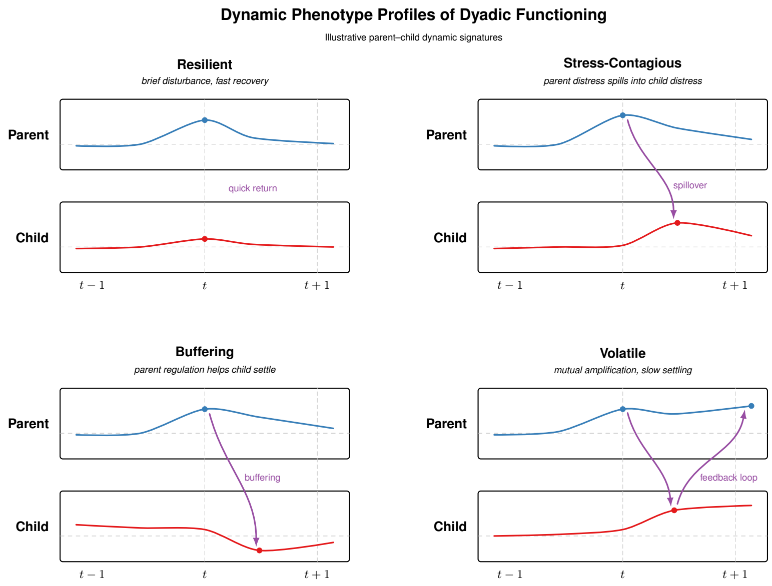
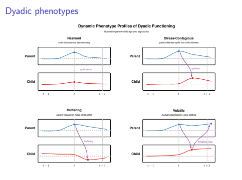
Question: How can we draw population-level conclusions from person-specific dynamics without averaging away the very individual differences we care about?

- Study 3 asks how to preserve **person-specific dynamics** while still making valid **population-level** inferences.
- For me, this is not only a statistical problem.
- It is the problem behind many HDFS questions, because we often care about **dynamic parameters** that capture regulation, adaptation, spillover, and risk in ways that **differ meaningfully across people, dyads, and families**.

Dyadic phenotypes



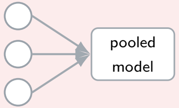


- Before I show the mechanics of MetaVAR, I want to show the kind of scientific target I care about.
- Imagine estimating a dynamic signature for each parent–child dyad: some dyads recover quickly after disturbance, some show stress contagion, some show buffering, and some show mutual amplification.
- Those are not just abstract parameters. They are candidate dynamic phenotypes of dyadic functioning.
- And once we can estimate those signatures well, the next question is **how to synthesize them without flattening the very heterogeneity that makes them interesting**.
- That is the problem MetaVAR is designed to solve.

The trade-off MetaVAR is designed to solve

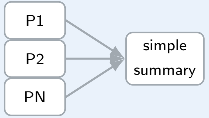
Population inference | **Idiographic fidelity**

One-step hierarchical models



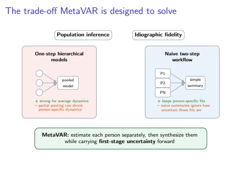
+ strong for average dynamics
− partial pooling can shrink person-specific dynamics

Naive two-step workflow



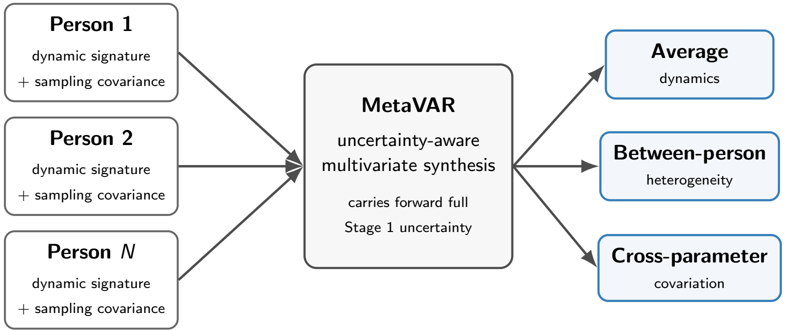
+ keeps person-specific fits
− naive summaries ignore how uncertain those fits are

MetaVAR: estimate each person separately, then synthesize them while carrying **first-stage uncertainty** forward



- One-step hierarchical models are attractive when you want population inference, because they estimate the average and individual differences together. But the trade-off is that person-specific dynamics get pulled toward the population mean.
- At the other extreme, you can fit a model per person and summarize afterward. That keeps the idiographic fits and scales well, but those summaries can act as if every person-level estimate is equally precise.
- MetaVAR is the middle ground: keep the person-specific models, then carry their uncertainty into the second stage.
- In HDFS terms, this matters whenever we want to say not only what the average dyad or person looks like, but how much meaningful variation there is around that average and what predicts it.

How MetaVAR works



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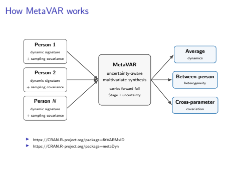
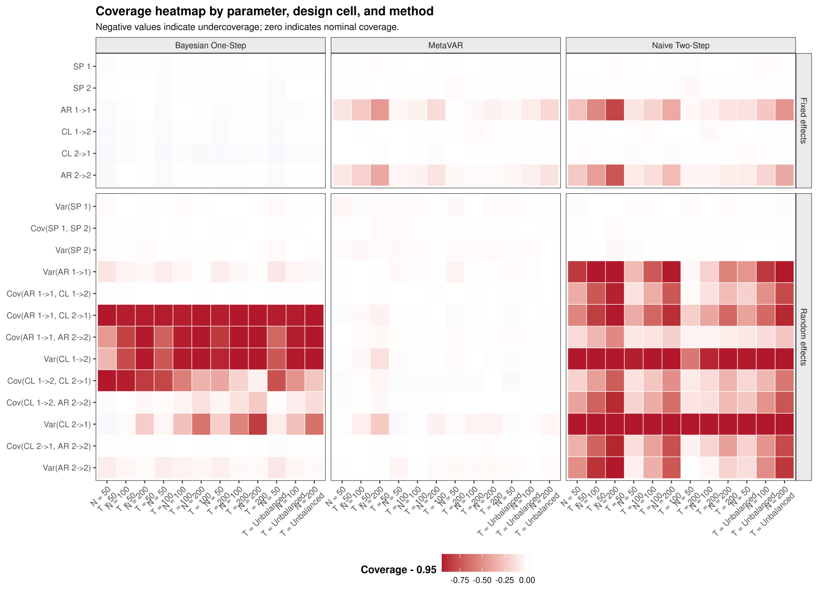
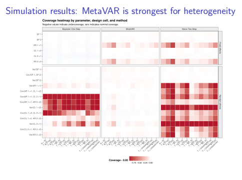
- https://CRAN.R-project.org/package=metaDyn



- Stage 1 is person-specific estimation: each individual gets an idiographic model, which yields a dynamic signature and its sampling covariance matrix.
- Stage 2 is MetaVAR: instead of treating those person-specific estimates as error-free, the method synthesizes both the signatures and their uncertainty.
- That gives us valid population-level inference on average dynamics, between-person heterogeneity, and cross-parameter covariation.

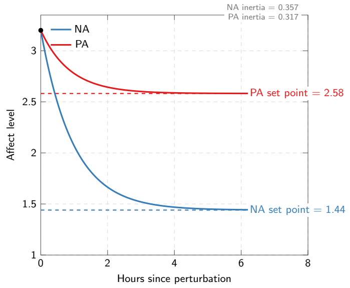
Simulation results: MetaVAR is strongest for heterogeneity

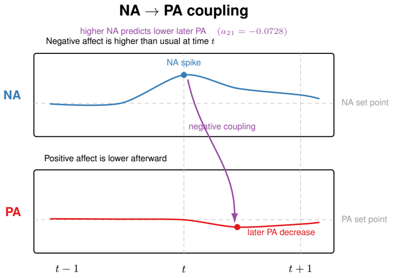


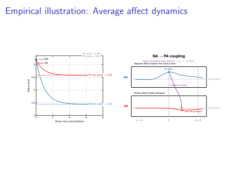


- The simulation tells me where MetaVAR is most useful. I do not present it as a universal replacement for one-step models. For population averages, the Bayesian multilevel comparator stayed competitive. MetaVAR's clearest advantage was when the scientific target was heterogeneity: how much people differ, and how well we calibrate uncertainty about those differences.

Empirical illustration: Average affect dynamics

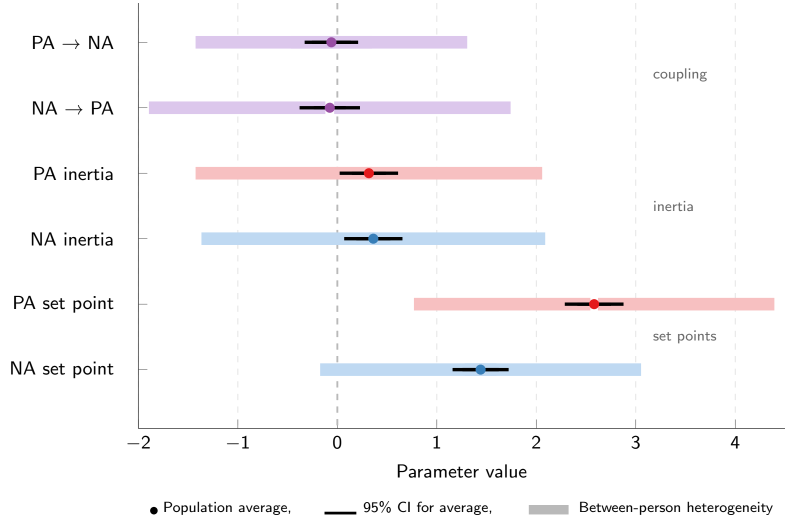


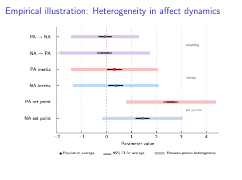




- The average affect system is interpretable: on average, people show higher positive affect than negative affect, and both processes show positive inertia.
- I am also showing one representative cross-lagged result here: higher negative affect predicts lower later positive affect.
- The companion result goes in the same direction for positive affect predicting lower later negative affect, so the average system reflects reciprocal inhibitory coupling.
- So MetaVAR gives us an interpretable population-level affect system, not just a bag of person-specific estimates.

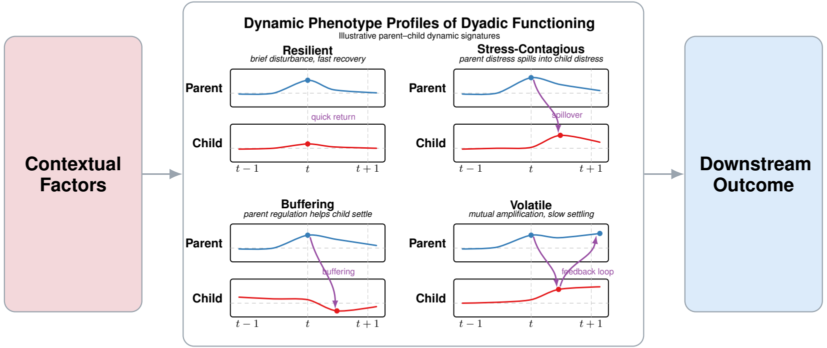
Empirical illustration: Heterogeneity in affect dynamics



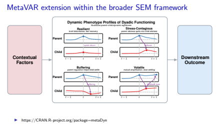


- The more important result is not only where the average lies, but how much people differ around it.
- The dot is the population average, the thin interval is the 95% confidence interval for that average, and the wider band shows approximate between-person heterogeneity around it.
- More specifically, that wider band comes from the random-effects variance for each parameter, so it gives a population spread rather than uncertainty about the mean.
- So people differ meaningfully in their set points, in how persistent affect is, and in how strongly negative and positive affect influence one another over time.
- The wider band is an approximate between-person interval, based on the random-effects variance for that parameter under a normal random-effects approximation.
- The empirical payoff here is that heterogeneity stops being a

MetaVAR extension within the broader SEM framework



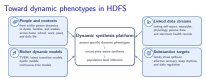
- https://CRAN.R-project.org/package=metaDyn



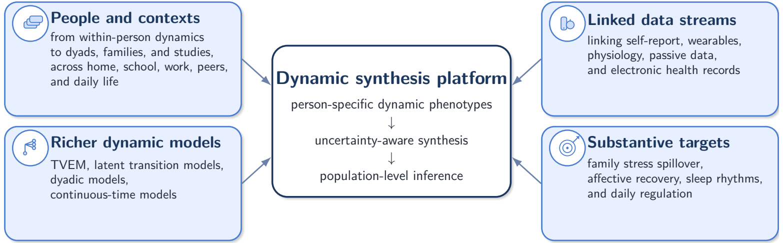
- This slide shows why MetaVAR matters beyond one methods paper.
- In a broader SEM framework, person-specific dynamic signatures can be modeled as outcomes of contextual factors and as predictors of later outcomes.
- So instead of stopping at “people differ,” we can ask what predicts those differences and why they matter downstream.
- That is the bridge to the next slide: toward dynamic phenotypes in HDFS.

Future directions

Future directions

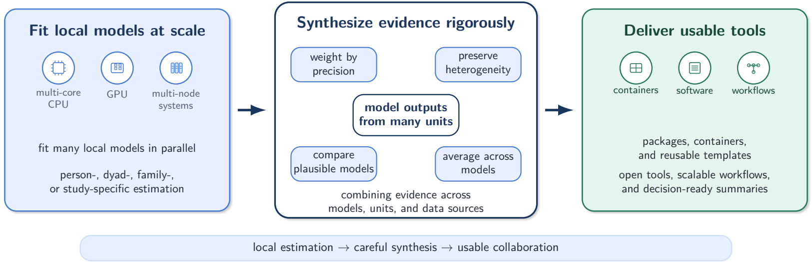


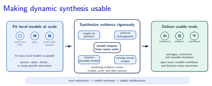
Toward dynamic phenotypes in HDFS



- This slide is where I see the work going next.
- The core idea is to estimate dynamic phenotypes, carry their uncertainty forward, and make valid population-level inferences about them.
- The people-and-contexts panel reflects the multilevel structure of the questions I want to study in HDFS—from persons to dyads, families, studies, and everyday settings.
- The linked-data panel reflects the kinds of data that increasingly define the field.
- The richer-models panel reflects the model families that can support those questions.
- And the substantive-targets panel keeps the agenda grounded in family life, development, and health.
- So the goal is not methods for their own sake. It is to help HDFS researchers study meaningful dynamic phenotypes in ways that are statistically principled and practically usable.

Making dynamic synthesis usable





- The infrastructure piece is important to me, but I want to frame it in service of collaboration rather than computation for its own sake.
- Many HDFS projects are naturally modular: person-, dyad-, family-, or study-specific models can often be estimated separately and in parallel.
- The harder question is how to combine those results well without pretending every estimate is equally precise.
- And the practical question is how to turn that into workflows colleagues and students can actually use: packages, templates, containers, and training materials.
- So again, the through-line is the same: methods matter more when they become usable for collaborative science.

Addressing Mechanisms and Heterogeneity in Change with Usable Tools

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Prevention and Methodology Training Program (PAMT)
Edna Bennett Pierce Prevention Research Center

The Pennsylvania State University

April 20, 2026

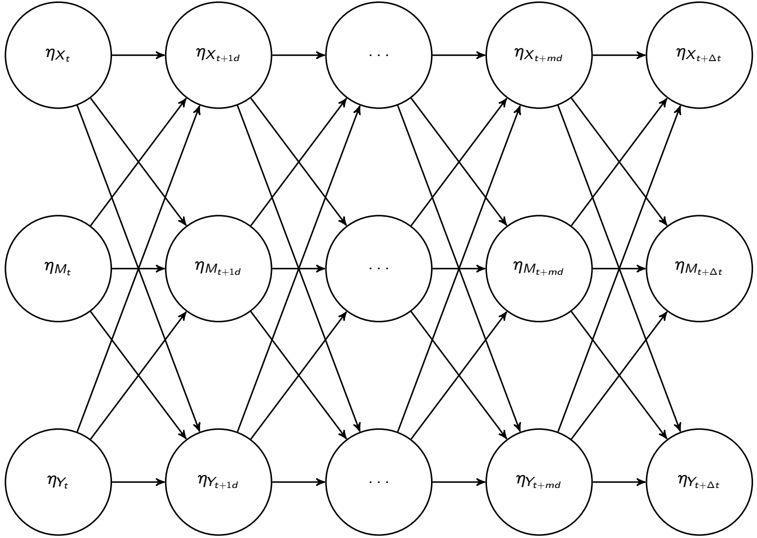
https://github.com/jeksterslab/talks20260420PSUHDFSJobtalk

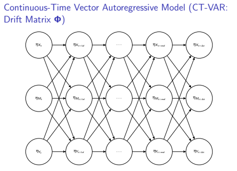
Addressing Mechanisms and Heterogeneity in Change with Usable Tools

- Thank you very much for your attention.
- I've tried to show a research program centered on dynamic processes: how change unfolds over time, how people and relationships differ in those processes, and how we can build tools that make those questions easier to study well.
- What excites me most about HDFS is that these are exactly the kinds of questions the field asks — in development, family life, health, and daily contexts.
- I would be excited to contribute here not only through methods development, but through collaboration, student training, and building usable quantitative infrastructure for substantive research.
- Thank you, and I'd be very happy to take questions.

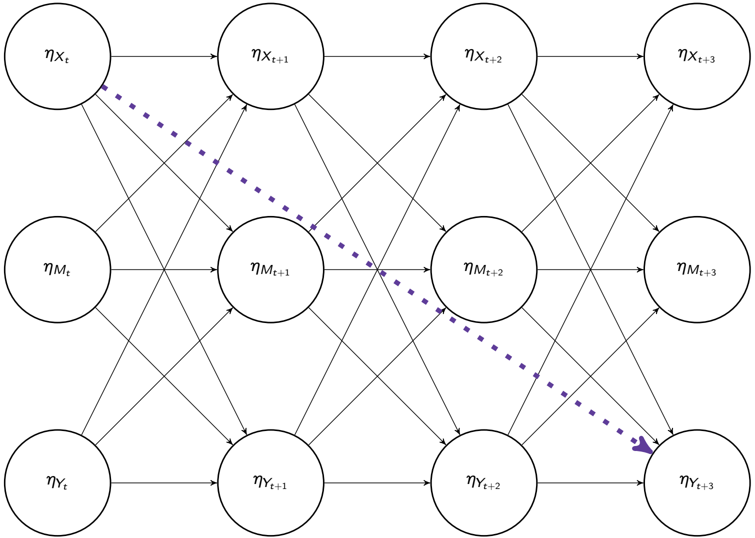
Extra slides

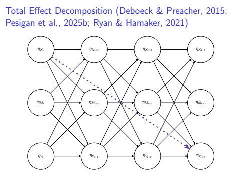
Continuous-Time Vector Autoregressive Model (CT-VAR: Drift Matrix $\mathbf{\Phi}$)





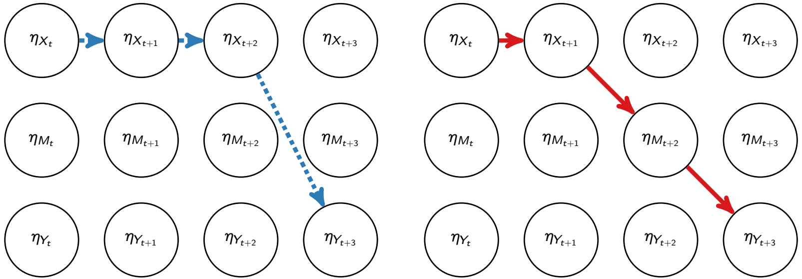
Total Effect Decomposition (Deboeck & Preacher, 2015; Pesigan et al., 2025b; Ryan & Hamaker, 2021)

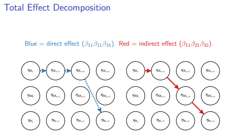




Total Effect Decomposition

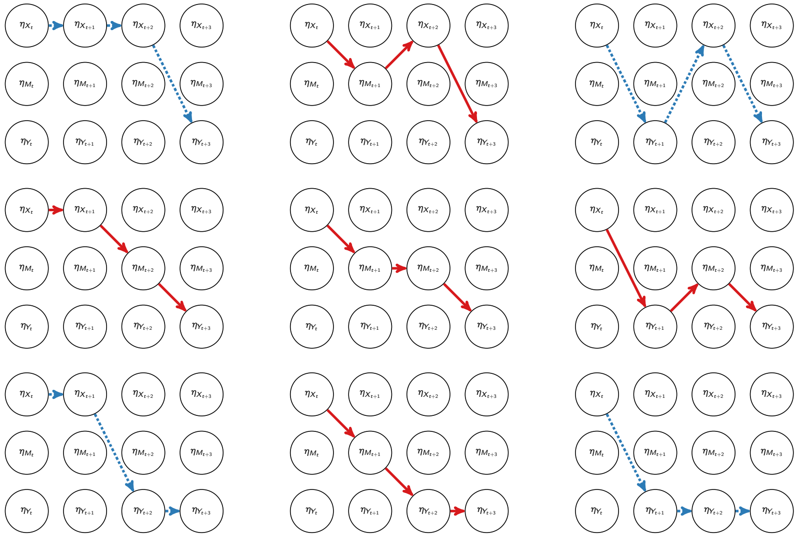
Blue = direct effect ($\beta_{11}\beta_{11}\beta_{31}$). Red = indirect effect ($\beta_{11}\beta_{21}\beta_{32}$).

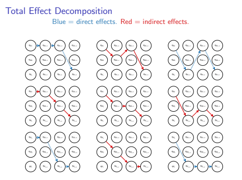




Total Effect Decomposition

Blue = direct effects. Red = indirect effects.





Total Effect Decomposition

Total, Direct, and Indirect Effects:

$$\text{Total} = \exp\left(\Delta t \boldsymbol{\Phi}\right) = \boldsymbol{\beta}_{\Delta t} \tag{1}$$

$$\text{Direct} = \exp\left(\Delta t \mathbf{D_m} \boldsymbol{\Phi} \mathbf{D_m}\right) \tag{2}$$

$$\text{Indirect} = \text{Total} - \text{Direct}. \tag{3}$$

Standardized Effects: Effect Size Measures

$$\text{Total}^{*}_{i,j} = \text{Total}_{i,j} \left(\frac{\sigma_{x_j}}{\sigma_{y_i}} \right), \tag{4}$$

$$\text{Direct}^{*}_{i,j} = \text{Direct}_{i,j} \left(\frac{\sigma_{x_j}}{\sigma_{y_i}} \right), \qquad \text{and} \tag{5}$$

$$\text{Indirect}^{*}_{i,j} = \text{Total}^{*}_{i,j} - \text{Direct}^{*}_{i,j} \tag{6}$$

Total Effect Decomposition
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CT-VAR (Chow et al., 2023; Deboeck & Preacher, 2015)

Measurement component:

$$\mathbf{y}_{i,t_m} = \boldsymbol{\nu} + \boldsymbol{\Lambda}\boldsymbol{\eta}_{i,t_m} + \boldsymbol{\varepsilon}_{i,t_m}, \boldsymbol{\varepsilon}_{i,t_m} \sim \mathcal{N}\left(\mathbf{0}, \boldsymbol{\Theta}\right) \tag{7}$$

Dynamic component:

$$\mathrm{d}\boldsymbol{\eta}_{i,t} = \left(\color{red}{\boldsymbol{\Phi}}\boldsymbol{\eta}_{i,t}\right)\mathrm{d}t + \boldsymbol{\Sigma}^{\frac{1}{2}}\mathrm{d}\mathbf{W}_{i,t} \tag{8}$$

Three-variable CT-VAR mediation model:

$$\mathbf{y} = \begin{pmatrix} X \\ M \\ Y \end{pmatrix}, \quad \text{and} \quad \boldsymbol{\eta} = \begin{pmatrix} \eta_X \\ \eta_M \\ \eta_Y \end{pmatrix} \tag{9}$$

Integral form:

$$\boldsymbol{\eta}_{i,t_{m_i}} = \boldsymbol{\alpha}_{\Delta t_{m_i}} + \color{red}{\boldsymbol{\beta}_{\Delta t_{m_i}}}\boldsymbol{\eta}_{i,t_{m_i-1}} + \boldsymbol{\zeta}_{i,t_{m_i}}, \boldsymbol{\zeta}_{i,t_{m_i}} \sim \mathcal{N}\left(\mathbf{0}, \boldsymbol{\Psi}_{\Delta t_{m_i}}\right) \tag{10}$$

$$\color{red}{\boldsymbol{\beta}_{\Delta t_{m_i}} = \exp\left(\Delta t \boldsymbol{\Phi}\right)} \tag{11}$$

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Delta Method Confidence Intervals

We define the delta method following the multivariate central limit theorem. Let $\boldsymbol{\theta}$ be $\mathrm{vec}\left(\boldsymbol{\Phi}\right)$, that is, the elements of the drift matrix $\boldsymbol{\Phi}$ in vector form sorted column-wise. Let $\hat{\boldsymbol{\theta}}$ be $\mathrm{vec}\left(\hat{\boldsymbol{\Phi}}\right)$, that is, the estimates of the elements of the drift matrix. The uncertainty around any functions of the estimates for $\hat{\boldsymbol{\theta}}$ is assumed to be characterized as:

$$\sqrt{n}\left(\mathbf{g}\left(\hat{\boldsymbol{\theta}}\right) - \mathbf{g}\left(\boldsymbol{\theta}\right)\right) \xrightarrow{\mathrm{D}} \mathcal{N}\left(0, \mathbf{J}\boldsymbol{\Gamma}\mathbf{J}'\right) \tag{12}$$

where $\mathbf{g}$ is a vector of functions, $\mathbf{J}$ is the matrix of first-order derivatives of the functions in $\mathbf{g}$ with respect to the elements of $\boldsymbol{\theta}$ and $\boldsymbol{\Gamma}$ is the asymptotic variance-covariance matrix of $\hat{\boldsymbol{\theta}}$. The approximate distribution of $\mathbf{g}\left(\hat{\boldsymbol{\theta}}\right)$ is given by

$$\mathbf{g}\left(\hat{\boldsymbol{\theta}}\right) \approx \mathcal{N}\left(\mathbf{g}\left(\boldsymbol{\theta}\right), n^{-1}\mathbf{J}\boldsymbol{\Gamma}\mathbf{J}'\right). \tag{13}$$

When $\boldsymbol{\Gamma}$ is unknown, by substitution, we can use the estimated sampling variance-covariance matrix of $\hat{\boldsymbol{\theta}}$, that is, $\hat{\mathbb{V}}\left(\hat{\boldsymbol{\theta}}\right)$ for $n^{-1}\boldsymbol{\Gamma}$.

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Monte Carlo (MC) Method Confidence Intervals

Based on the asymptotic properties of maximum likelihood estimators, we can assume that they are normally distributed around the population parameters

$$\hat{\boldsymbol{\theta}} \sim \mathcal{N}\left(\boldsymbol{\theta}, \mathbb{V}\left(\hat{\boldsymbol{\theta}}\right)\right). \tag{14}$$

Using this distributional assumption, a sampling distribution of $\hat{\boldsymbol{\theta}}$ which we refer to as $\hat{\boldsymbol{\theta}}^{*}$ can be generated by replacing the population parameters with sample estimates, that is,

$$\hat{\boldsymbol{\theta}}^{*} \sim \mathcal{N}\left(\hat{\boldsymbol{\theta}}, \hat{\mathbb{V}}\left(\hat{\boldsymbol{\theta}}\right)\right). \tag{15}$$

A sampling distribution of $\mathbf{g}\left(\hat{\boldsymbol{\theta}}\right)$, which we refer to as $\mathbf{g}\left(\hat{\boldsymbol{\theta}}^{*}\right)$, can be generated by using the simulated estimates to calculate $\mathbf{g}$. The standard deviations of the simulated estimates are the SEs. In resampling approaches, it is customary to use the percentiles of the generated distribution as CIs instead of the SEs.

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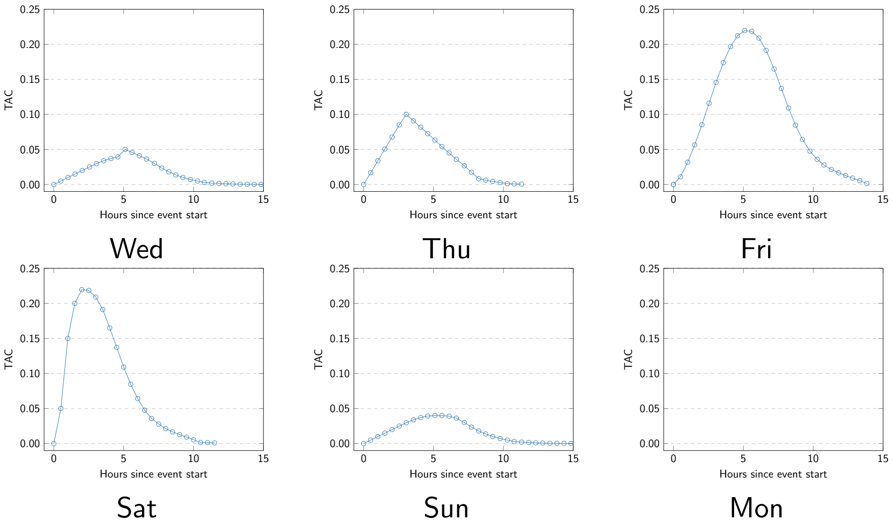
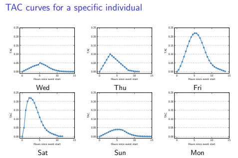
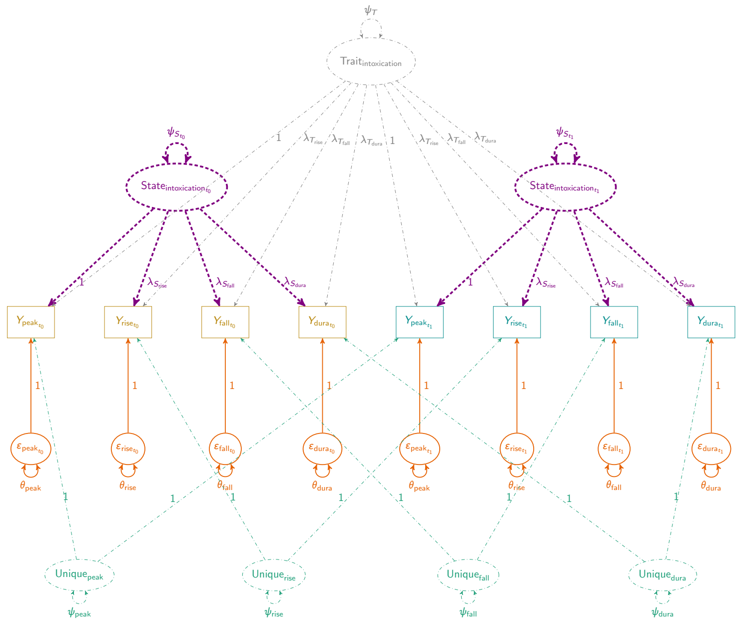
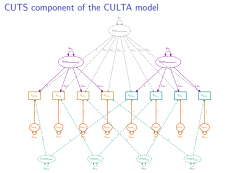
TAC curves for a specific individual



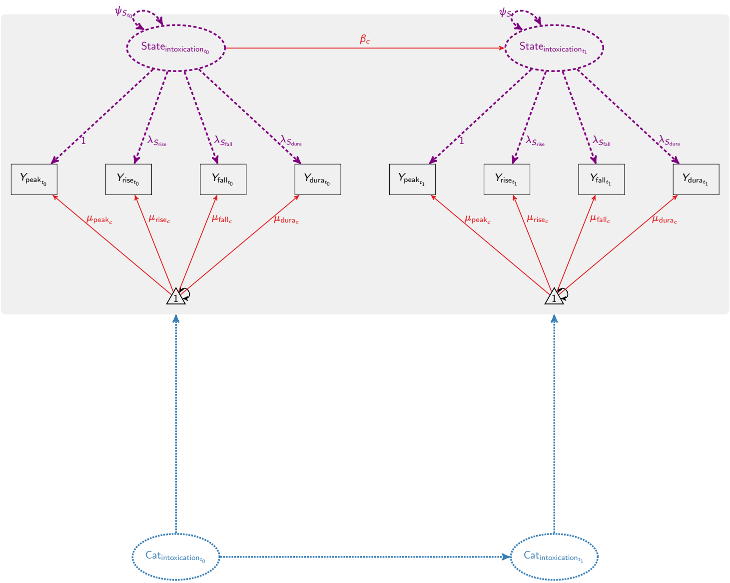


CUTS component of the CULTA model





LTA component of the CULTA model

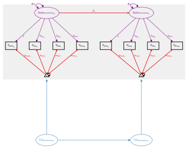


2026-04-16

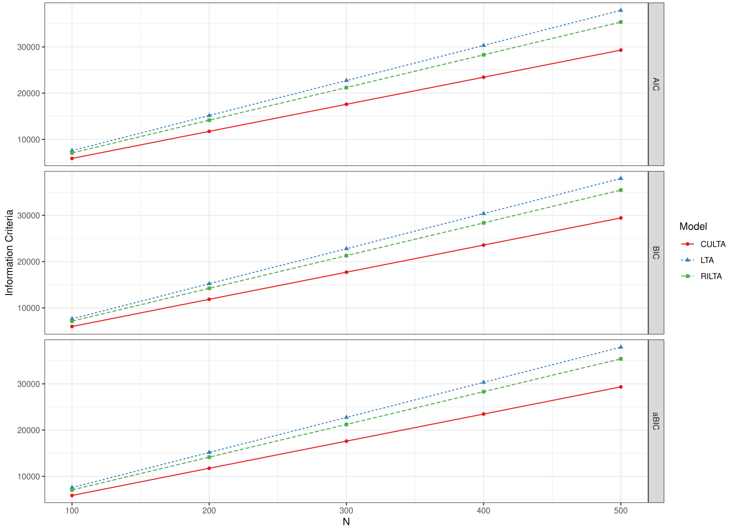
Addressing Mechanisms and Heterogeneity in Change with Usable Tools

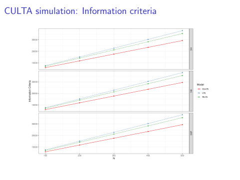
└Extra slides

└LTA component of the CULTA model

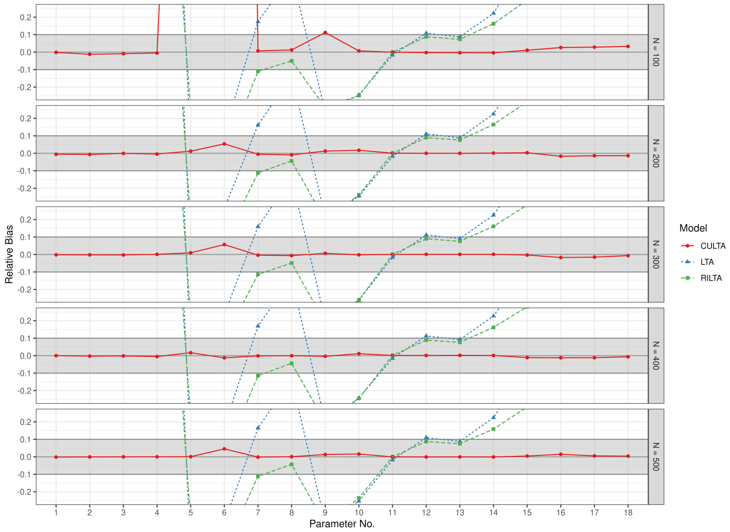


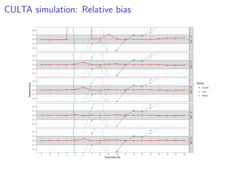
CULTA simulation: Information criteria



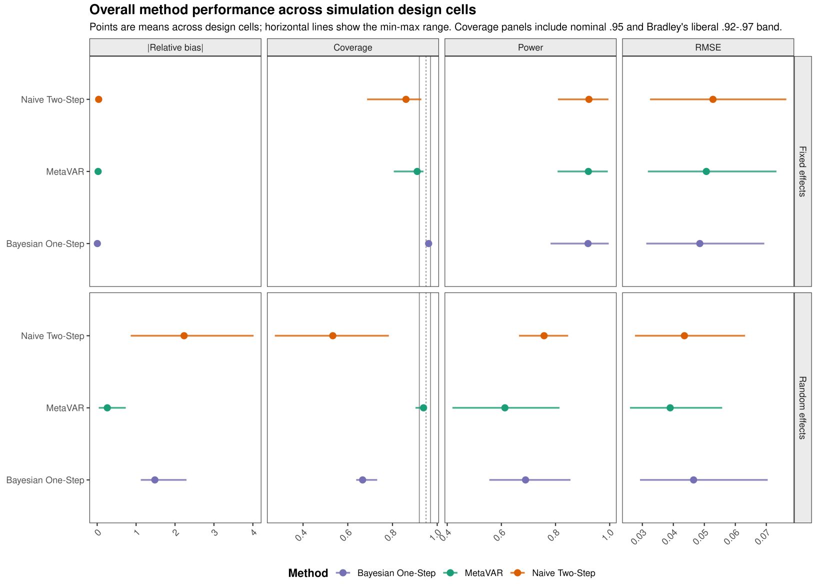


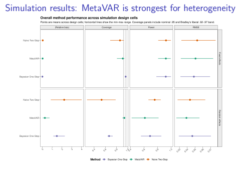
CULTA simulation: Relative bias





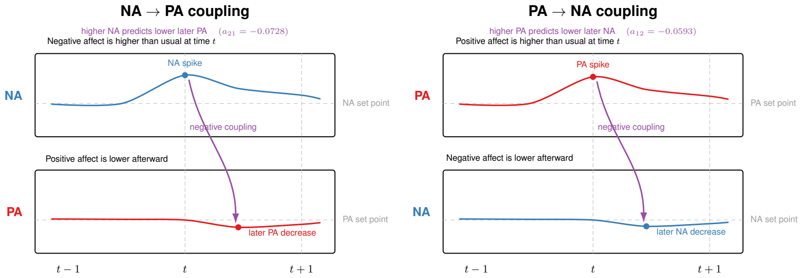
Simulation results: MetaVAR is strongest for heterogeneity

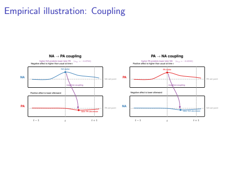




- The simulation tells me where MetaVAR is most useful. I do not present it as a universal replacement for one-step models. For population averages, the Bayesian multilevel comparator stayed competitive. MetaVAR's clearest advantage was when the scientific target was heterogeneity: how much people differ, and how well we calibrate uncertainty about those differences.

Empirical illustration: Coupling





References

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